

Name: _____

Date: _____

VOLUME AND SURFACE AREA, (VOLUME OF PYRAMIDS, CONES AND SPHERES)

LO: I can calculate the volume of pyramids, cones and spheres using standard formulae.

Volume of pyramids, cones and spheres

A **pyramid** has a polygon base and triangular faces meeting at an apex. A **cone** is a circular pyramid. Their volumes are one-third of the corresponding prism or cylinder.

Key formulae: Pyramid: $V = \frac{1}{3} \times \text{base area} \times \text{height}$. Cone: $V = \frac{1}{3}\pi r^2 h$. Sphere: $V = \frac{4}{3}\pi r^3$.

Quick examples

●	Pyramid: $V = \frac{1}{3} \times \text{base area} \times h$	one-third of the prism with same base
●	Cone: $V = \frac{1}{3}\pi r^2 h$	one-third of the cylinder
●	Sphere: $V = \frac{4}{3}\pi r^3$	uses radius cubed
●	Cone $V = \pi r^2 h$	must include the $\frac{1}{3}$ factor
●	Sphere $V = 4\pi r^2$	that is surface area, not volume

Key idea

Pyramids and cones have volume equal to one-third of the corresponding prism or cylinder. A sphere uses its own formula with r cubed.

$$V = \left(\frac{1}{3}\right) \times \text{base area} \times h$$

The $\frac{1}{3}$ factor applies to all pyramids and cones

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – square-based pyramid

Find the volume of a pyramid with a square base of side 6 cm and height 10 cm.

$$\text{Base area} = 6 \times 6 = 36 \text{ cm}^2$$

$$V = \frac{1}{3} \times 36 \times 10 \\ = 120 \text{ cm}^3$$

$$\boxed{120 \text{ cm}^3}$$

Find the base area first, then multiply by one-third of the height.

Example 2 – cone

Find the volume of a cone with radius 5 cm and height 12 cm. Round to 1 d.p.

$$V = \frac{1}{3}\pi r^2 h \\ = \frac{1}{3} \times \pi \times 25 \times 12 \\ = 100\pi = 314.2 \text{ cm}^3$$

$$\boxed{314.2 \text{ cm}^3}$$

Substitute into the cone formula and evaluate.

Example 3 – sphere

Find the volume of a sphere with radius 3 cm. Round to 1 d.p.

$$V = \frac{4}{3}\pi r^3 \\ = \frac{4}{3} \times \pi \times 27 \\ = 36\pi = 113.1 \text{ cm}^3$$

$$\boxed{113.1 \text{ cm}^3}$$

Cube the radius, then multiply by $\frac{4}{3}\pi$.

Example 4 – hemisphere

Find the volume of a hemisphere with radius 6 cm. Round to 1 d.p.

$$\text{Full sphere: } V = \frac{4}{3}\pi \times 6^3 = \frac{4}{3}\pi \times 216 \\ = 288\pi$$

$$\text{Hemisphere} = \text{half: } \frac{1}{2} \times 288\pi = \\ 144\pi \\ = 452.4 \text{ cm}^3$$

$$\boxed{452.4 \text{ cm}^3}$$

Find the full sphere volume, then halve it.

YOUR TURN – PRACTICE (deliberate practice)

1. Pyramids

Find the volume of each pyramid.

- a) Square base side 4 cm, height 9 cm
- b) Square base side 10 cm, height 12 cm
- c) Rectangular base 6×8 cm, height 15 cm
- d) Triangular base area 20 cm^2 , height 9 cm

2. Cones

Find the volume of each cone. Round to 1 d.p.

- a) $r = 3 \text{ cm}$, $h = 7 \text{ cm}$
- b) $r = 4 \text{ cm}$, $h = 9 \text{ cm}$
- c) $r = 6 \text{ cm}$, $h = 10 \text{ cm}$
- d) $r = 5 \text{ cm}$, $h = 15 \text{ cm}$

3. Spheres

Find the volume of each sphere. Round to 1 d.p.

- a) $r = 4 \text{ cm}$
- b) $r = 6 \text{ cm}$
- c) $r = 10 \text{ cm}$
- d) Diameter 14 cm

4. Hemispheres and composite shapes

Find the volume. Round to 1 d.p.

- a) Hemisphere: $r = 5 \text{ cm}$
- b) Hemisphere: $r = 8 \text{ cm}$
- c) Cone on top of hemisphere, both $r = 3 \text{ cm}$, cone $h = 4 \text{ cm}$
- d) Cylinder $r = 4$, $h = 10$ plus hemisphere $r = 4$ on top

5. Mixed volume problems

Solve each problem. Round to 1 d.p. where needed.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Volume of a cone = $\frac{1}{3}\pi r^2 h$ ☐

Correct answer: _____

Reason: _____

b) Volume of a cone = $\pi r^2 h$ ☐

Correct answer: _____

Reason: _____

c) Volume of a sphere uses r^3 (radius cubed) ☐

Correct answer: _____

Reason: _____

d) Doubling the radius of a sphere doubles its volume ☐

Correct answer: _____

Reason: _____

e) A hemisphere volume is half of $\frac{4}{3}\pi r^3$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) An ice cream cone has radius **3 cm** and height **12 cm**. A hemisphere of ice cream (same radius) sits on top. Find the total volume of cone plus ice cream. If the ice cream melts into the cone, will it overflow? Show your working.

Expression: _____

Simplified: _____

b) A solid metal cone has base radius **6 cm** and height **10 cm**. It is melted down and recast into a sphere. Find the radius of the sphere. Give your answer to 1 decimal place.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

